

Roll No.

--	--	--	--	--	--	--

Paper Code: TCS 302

Mid Semester Examination 2022

B.Tech (CSE/CE/CST/AI-ML/AI-DS) III Semester

Data Structure with 'C' language.

Time: 1:30 Hrs

MM: 50

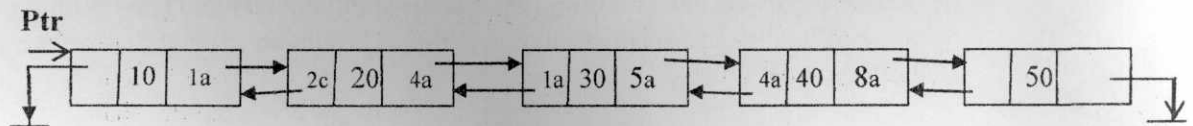
INSTRUCTIONS TO STUDENTS

Note :

- (i) This question paper contains five questions with alternative choice.
- (ii) All questions are compulsory.
- (iii) Each question carries **two** parts **a** or **b**. Attempt either parts **a** or **b** of each question.
- (iv) Total marks assigned to each question are ten.

Q1.

- A) Assume that you have a double linked list, first node of the linked list is pointed by pointer Ptr, write a C function to delete the last node of the linked list. (10)



OR

- B) Assume that you have two single linked lists. First of the first linked list is pointed by a pointer P and first node of second linked list is pointed by pointer Q. Write a C function to concatenate the second linked list after first linked list. (10)

Q2.

- A) What do you mean by a dynamic array? Write a 'C' function to create a dynamic array to store N elements and then find and print 4th repeating element in the array. (10)

OR

- B) Assume that you a stack implemented with linked list, write a C function to print the stack in reverse order (i.e. from bottom to top, do not use array). (10)

Q3.

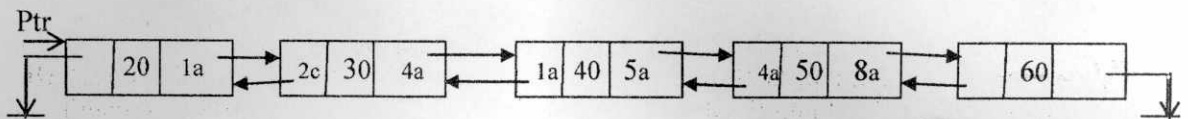
- A) Assume that you have a single linked list; first node of the linked list is pointed by a pointer PTR. Write a C function to delete alternate nodes of the linked list. (10)

OR

- B) Assume that you have a single linked list with a pointer P, at first node. Write a C function to input a number and search it in the linked list if number is found update the linked list by inserting a new node after the node for which key is found otherwise print number not found. (10)

Q4.

- A) Assume that you have a double linked list, first node of the linked list is pointed by pointer Ptr, write a C function to insert a new node after first node in the linked list. (10)

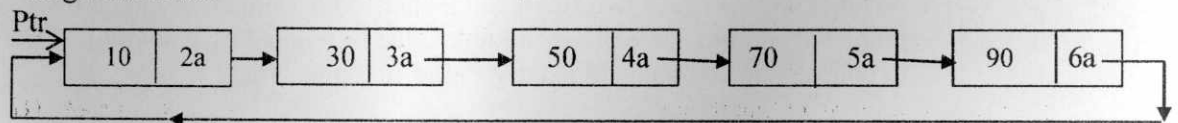


OR

- B) Assume that you have two singly linked lists. Pointers P and Q are pointing to first node of the linked lists respectively. Write a C function to print dissimilar nodes from both the linked lists. (10)

Q5.

- A) Consider the following circular linked list, a pointer Ptr is pointing to the node which is inserted first in the linked list. Write a C function to convert the circular linked list into a single linked list. (10)



OR

- B) Assume that you have a single linked list; first node of the linked list is pointed by a pointer PTR. Write a C function to print the nodes in alternate order till last node. (10)